Software Engineering

CSC648/848

Henry
VILLAR

GUI PRINCIPLES

Design, Development and Evaluation

SW Engineering Process Components

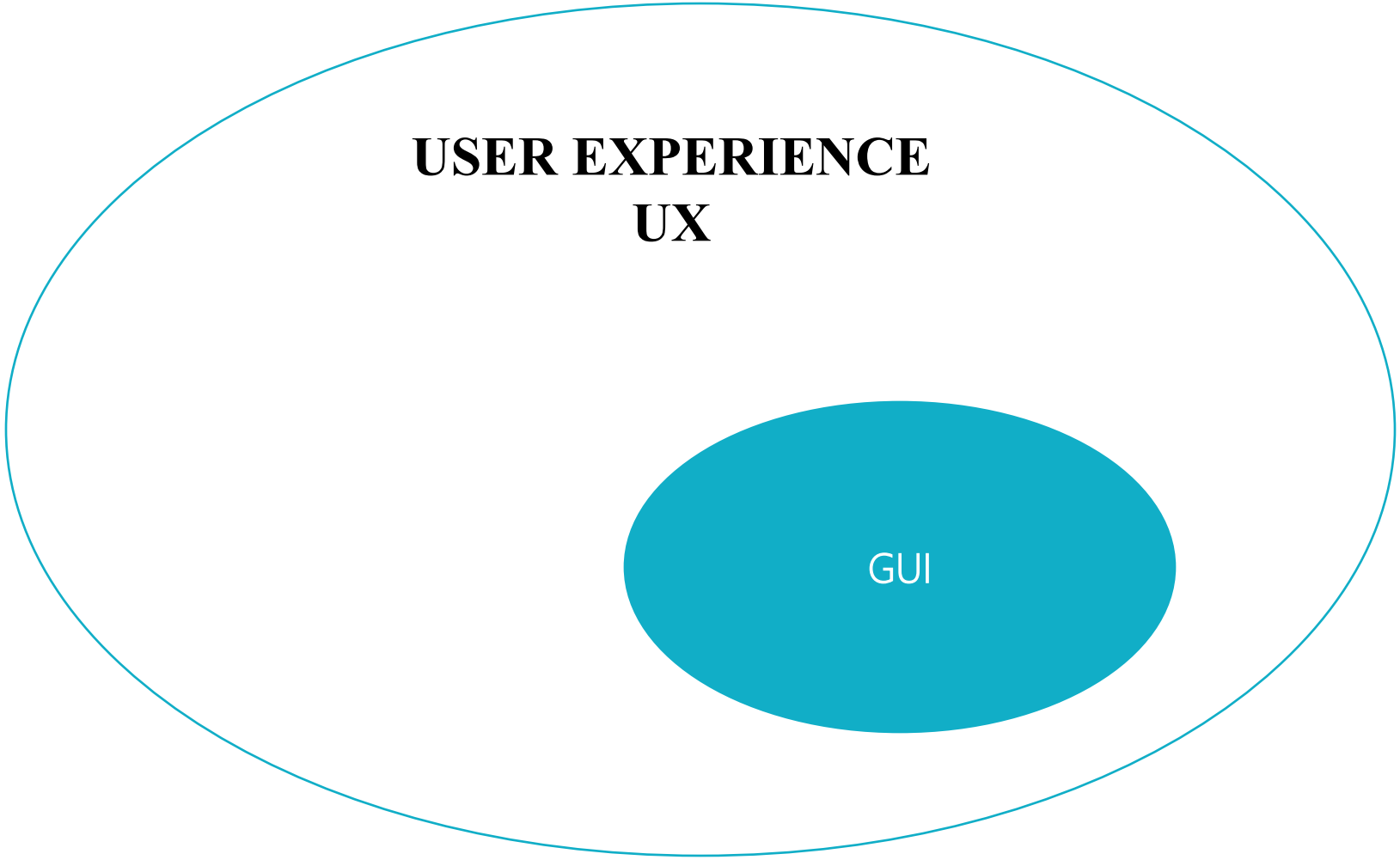
1. *Requirements and Specification*
2. *Design and Implementation*
3. *Validation*
4. *Maintenance and Evolution*

Usability & GUI Principles

- Usability can be described as the capacity of a system to provide a condition for its users to perform the tasks:
  - Safely
  - Effectively
  - Efficiently
- While enjoying the experience
- In **software engineering,** usability is the degree to which a software can be used by specified consumers to achieve quantified objectives with **effectiveness, efficiency, and satisfaction** in a quantified context of use.

- GUI Principles: General principles which guide UI design patterns
- Design Patterns: User Interface Design patterns are recurring solutions that solve common design problems. Design patterns are standard reference points for the experienced user interface designer.
- Development process: Personas>Functions>Mockups>horizontal prototype>coding
- UCD

User experience
Design


USER EXPERIENCE
UX
GUI

Important

User Experience – more than GUI

- *User experience* refers to ease of use of the product as a whole, and includes GUI but also everything else that "touches" the user:
  - Documentation
  - Installation
  - Purchase, registration
  - Support
  - API and SW tools usability
- For non-SW products:
  - Packaging
  - Assembly
  - Ergonomics (physical)
  - ….

Important

Your goal – total user satisfaction

- Make systems/sw that are all three:
  - **Useful:** A product which is useful is one that allows a user to accomplish a task or objective.
  - **Usable**. Usable refers to the usability of a given product. It is more than "useful" it examines the way that the product will be used and whether it enables the user to do so in a pleasurable, simple (or as simple as possible) and effective manner.
  - **Used.** A product may be both useful and usable and still fail to be used. The ultimate aim of a design is not to be useful or usable but for users to use that design. Without users a product is a failure and it doesn't matter how great the design was – it's still a failure.

Driven by users not SW engineers

Important

Always think of a USER

- Who will use my works (SW, app, also documents - recall M submission instructions…)
- What do they expect/know
- What are their "characteristics" and background
- What is the context of the usage
- What will they want to do

Lead to personas and use cases then to functions and design

Important

Practical GUI development process

Iterate and get feedback from users

- Develop **personas** (typical users) and **use cases (OK to be a bit more detailed with UI design)**
- Define and prioritize **functions**
- Create paper **B&W mockups** (NO programming)
  - Review mockups with internal users or domain experts and developers (internal focus groups) – **revise revise revise with users; use GUI design principles**
- **Create (horizontal) prototype** – clickable UI no back end
  - Review mockups with users or domain experts and developers (interval focus groups) – **revise revise revise with users (focus groups) ; use GUI design principles and patterns**
- **Start coding**
  - Get **feedback, revise revise revise with users** (focus groups); use established frameworks and stick to GUI design principles
- Show to wider audience including limited external users

Use cases for UX and UI

- Can be a bit more details as necessary to guide UI design
- Can be “mini-use cases” focusing on specific UI widgets in detail (slider vs. round dial, VCR play buttons vs. regular buttons)
- Should be discussed and reviewed by users (off-line or focus groups)
- Use for evaluation of mockups and UI working code

User Characteristics

Important

Know your user, client, customer!!!!

- First, some general observations about the user
  - How people interact with computers
  - Human characteristics of relevance to GUI

Important

Important Human Characteristics related to GUI: *Perception*

Awareness and understanding of elements and objects through sight, sound…. Influenced in part by experience

- *Proximity:* objects relate to each other
- *Similarity:* similar objects should relate
- *Simple* designs are easier to remember
- *Closure, unity:* objects forming closed shape perceived as groups
- *Balance and symmetry:* vertical and horizontal
- Subjective perception of motion, image, video,audio quality
- *Context:* color backgrounds influence color perception etc.

➔Implications for GUI screen and element design (to be covered later)

➔Drives GUI design patterns

Important

Important Human Characteristics related to GUI: *Movement Control- Fitts' Law*

- Keyboard, mouse, trackball usage involves movement
- Fitt's Law (1954) is one of the rare exact laws related to usability:

Time-to-acquire-target = a + b*Log2(2A/W)

(A is distance, W is width of the target)

➔

- Large objects for important functions
- Group related buttons
- Pining action of the sides, top, bottom and corners of screens
- Avoid small clickable objects

Important

Important Human Characteristics related to GUI: *Interactivity*

Response Time	**User reaction and perception**
< 0.1 sec.	Instantaneous, no interruptions
< 1.0 sec.	Delay noticed, but no break in thought stream
> 10.sec	Interruption, user switches to another task

Important

User/task considerations (for categorizing and surveying users)

<u>Knowledge and Experience</u>

- Computer Literacy
- System experience
- Application experience
- Task experience
- Education
- Reading level
- Typing skill
- Native language or Culture

Important

User/task considerations (for categorizing and surveying users)

Job/task/Need

- Type of system use
- Frequency of use
- Task importance
- Task structure
- Social interactions
- Training
- Turnover rate
- Job category
- Lifestyle

Important

User/task considerations (for categorizing and surveying users)

<u>Psychological Characteristics re: job/task</u>

- Attitude
- Motivation
- Patience
- Expectations
- Stress level
- Cognitive style

Important

User/task considerations (for categorizing and surveying users)

<u>Physical characteristics</u>

- Age
- Gender
- Handedness
- Disabilities

Human interaction speeds

- Reading: paper 200 w/minute; computer 180 w/min
- Listening: 150 w/minute
- Typing (classic): 60 w/minute average
- Keyboard typing: transcripts 33 w/minute; composition 19 words/minute
- Hand printing: memorized text 31 w/minute; copying 22 w/minute
- Expects system response at better than 0.5 sec. To be felt fully interactive

UI Design patterns

- UI design patterns follow UI principles – they offer practical designs guidance (and sometimes code) for common problems (registration, sequences, search, forms…)
- Major companies issuing their design patterns and guidance with tools (Apple, Google…)
- Analogy to architectural patterns
- Design Patterns:
  - Getting Input (Forms, Explaining the process, community driven)
  - Navigation (Tabs, Jumping hierarchy, menus, content)
  - Dealing with Data (tables, formatting data, images, search)
  - Social (Reputation, interaction)

GUI Design Patterns – implementation issues

- UX Designers and vendors publish design patterns and tools
- Use and leverage them

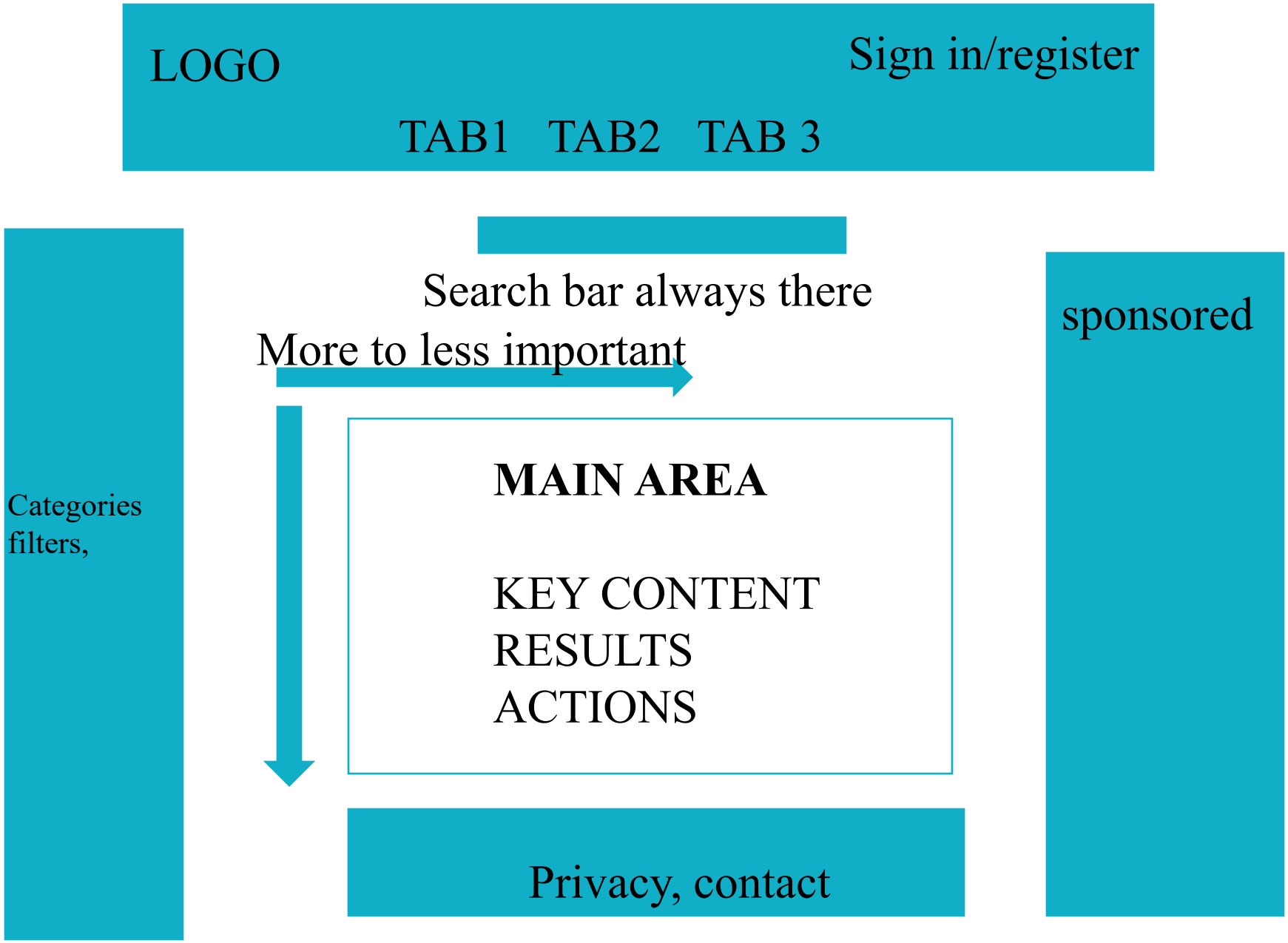
Apple and Google UI design principles and guidance → GUI design patterns and tools

- Apple
  - https://developer.apple.com/library/ios/documentation/UserExperience/Conceptual/MobileHIG/
  - https://developer.apple.com/design/tips/

- Google
  - http://developer.android.com/design/get-started/principles.html

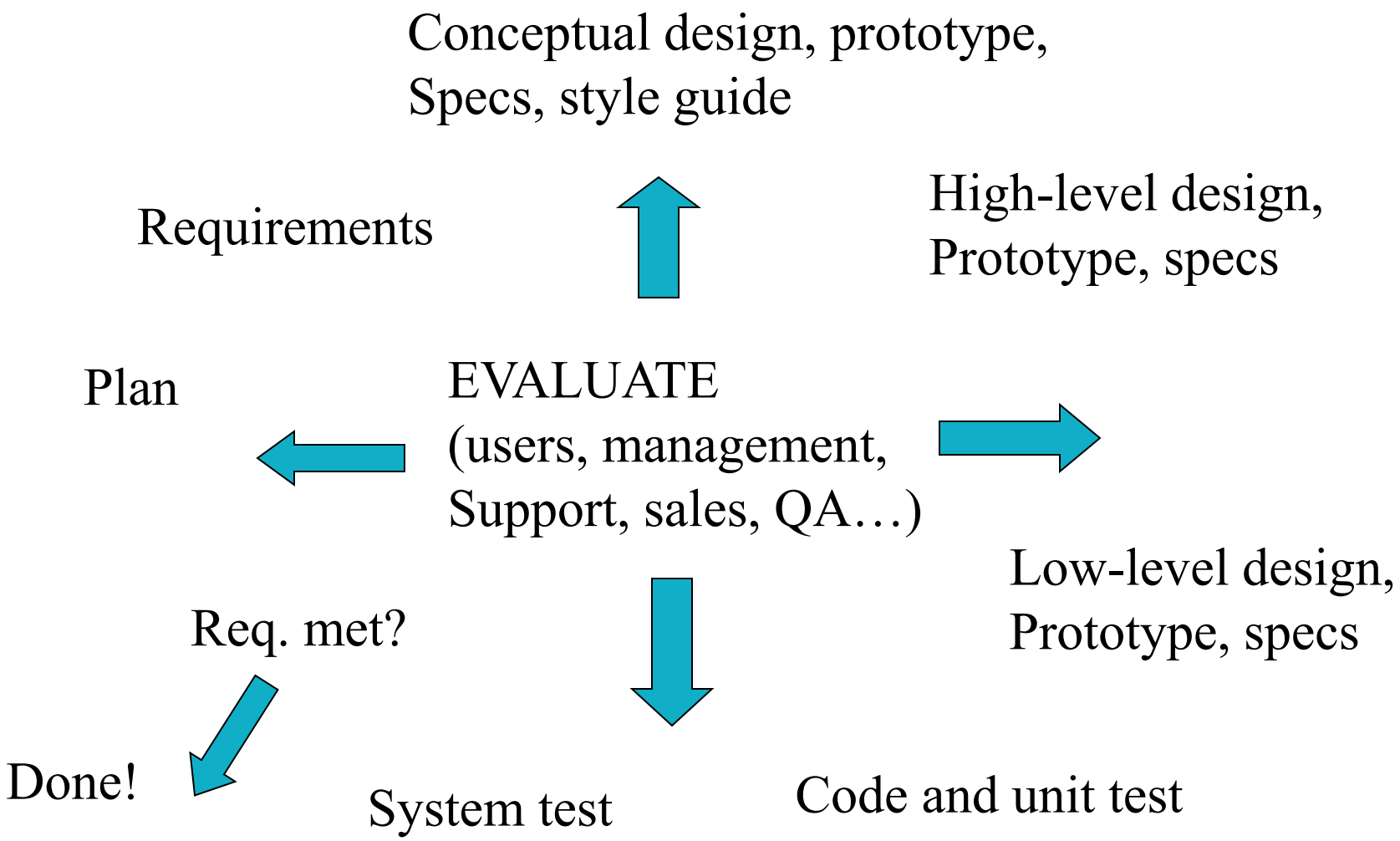
“Vanilla” WWW pages layout

Important



User-Centered Design and Development – MUST DO for GUI

Important



http://en.wikipedia.org/wiki/User-centered_design

High level Principles of User Interface Design

Important

- General principles which guide UI design patterns.
- Derived from inherent cognitive/perceptual human properties
- List given in alphabetical not priority order
- All principles are important, BUT *relative importance might depend on users and business objectives*

Have this as a check list to design/evaluate GUIs

How do you rank importance of GUI design principles?

- Aesthetically pleasing
- Clarity
- Compatibility
- Easy to understand
- Configurability
- Consistency
- Control
- Context
- Efficiency
- Familiarity
- Functionality
- Forgiveness, recovery
- Predictability
- Recovery
- Responsiveness
- Simplicity
- Transparency
- Design trade-offs

https://www.figma.com/resource-library/ui-design-principles/ (7)

My top:

- Simplicity
  - Control
  - Aesthetically pleasing
  - Functionality
  - Context

https://xd.adobe.com/ideas/process/ui-design/4-golden-rules-ui-design/

Important

Focus groups - Essential tool in developing requirements in UCD

- http://en.wikipedia.org/wiki/Focus_group
  - "In **usability engineering**, a focus group is a survey method to collect the views of users on software or a website. This marketing method can be applied to computer products to better understand the motivations of users and their perception of the product. Unlike other methods of ergonomics, focus group implies several participants: users or future users of the application. The focus group can only collect subjective data, not objective data on the use of the application as the usability test for example."
- Used for high level and also low level detailed feedback

How to run focus groups

Interaction Design Foundation (05/28/18)

- https://www.interaction-design.org/literature/article/how-to-conduct-focus-groups
- Choice of participants (recall your peronas)
- Preparing
- Conducting it
- Note taking
- Analysis

- The use and misuse…Nielsen Norman Group
  - https://www.nngroup.com/articles/focus-groups/

Important

Latest: UX re-design and evaluation in the era of on-line services and big data

- Products and services not released as structured downloads but continuously available and modified (e.g. on-line services, on-line search)
- UX testing evaluation and design is now augmented by "test-in-production" where UX is being continuously improved and evaluated based on real-time customer usage
  - Data scientists and ML experts analyze and discover/predict user patterns and behaviors ➔ products are redesigned

GUI Implementation issues

- To develop mockups use hand drawings or tools
- To code UI use tools like bootstrap

https://getbootstrap.com/

  - Offers widgets for everything
  - Cross browser compatibility
  - Responsiveness – adjusts to screen size and mobile UI
  - Forms patterns
  - Input field validation ➔ key to security

- Use automated tools for GUI QA
- Follow UCD process and iterate many times with key users

Important

GUI QA vs. GUI Usability evaluation

- QA testing
  - Find bugs
  - Browser compatibility
  - Test actual SW implementation
- Usability evaluation
  - How easy is it to use (bugs or not)
  - Can be done on mockups and prototypes and buggy SW

Important

UI design patterns for search

Search V GO

(*free text or KW*)

CATEGORY pull-down (driven from CATEGORY DB Table): limits search to e.g. books, music, movies

Search arguments (use some Fuzzy search like with %LIKE)

Animals
Things
Rockets
…..

GO

Categories (e.g. domain)

Search within chosen category

UI design pattern for search results”

Important

Back to Search

Showing 1-10 out of 12345

(*Your own back button; When back, keep old search arguments there*),

Thumbnail, preview

Image

Description here

More details

Order

Image

Description here

More details

Order

Key function

<< Previous 1 2 3 4 … Next >>

Context

(Try to show current search arg. on the same page)

More on UI for search

- Keep search arguments persistent – only user can change them
  - Why?
- If search input is a valid format (e.g. zip code) and no items found
  - Clarify no items found AND
  - In many applications like shopping, renting and search for a product or service it is preferable to offer alternatives (avoid blank screen with one error message)
- Do basic input validity check – if the search argument is NOT in correct format clarify this in error message
- Present search results in a way to be helpful to the user and answer most likely user goals (example: search for "Target Westgate Hours" – try different search engines and check format of the results)
- Design so search results are easy to understand and use on mobile (context: people are on the go and need search results quickly on mobile)
- Good UI design [patterns
  - https://uxmag.com/articles/designing-search-results-pages

Popular tools for testing PC UI subsystems (no endorsements, just some examples)

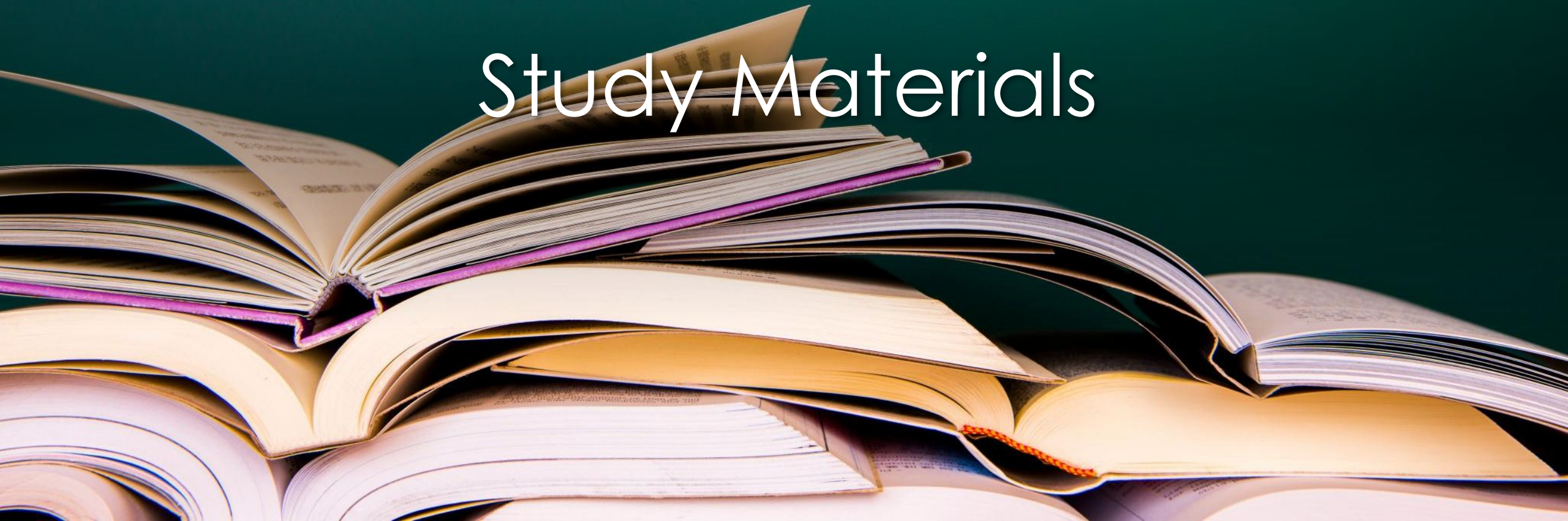
- Selenium http://seleniumhq.org/
- Qt Group
- https://www.qt.io/product/quality-assurance?gclid=CMGzuYWRpawCFQUaQgodTG1DDA
- Try it! Use it!
- Good for job search – many companies use it and require it..
- Some useful tutorials (there are many…)
  - http://www.baeldung.com/java-selenium-with-junit-and-testng
  - https://www.blazemeter.com/blog/how-to-run-a-selenium-test-with-junit
  - https://www.qt.io/resources/qt

Team Project Time

- Intro to M2 **More Detailed Requirements, Specs, Architecture, UI mock-ups and Vertical SW prototype**. Two goals:
  - ***More detailed design - Milestone 2 document***: prioritize requirements; design high level UI mockups and storyboards; design high level DB organization, and practice UML. *(one per team, submitted similarly like Milestone 1)*. Use as much space as you need, but the expected length is about 20-25 pages. You must submit <u>all</u> sections.
    1. Functional Requirements – prioritized
    2. <u>UI Mockups and Storyboards (high level only)</u>
    3. High level Architecture, Database Organization
    4. High Level UML Diagrams
    5. <u>Identify *actual* key risks for your project at this time</u>
    6. <u>Project management</u>

  - ***Vertical SW Prototype***: Develop first simple bare-bones prototype (from DB to browser) to test the infrastructure, educate the team, resolve technical issues and also serve as basis of further development (must run on the deployment server). *(one per team) to be reviewed in class and also by Class CTO (off-line)*

Study Materials

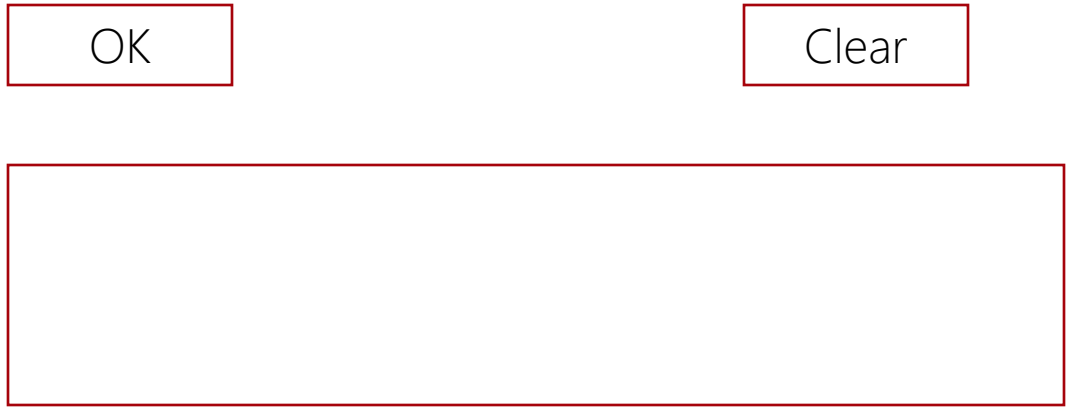


From R. Acosta former student

Examples (good and bad) UI

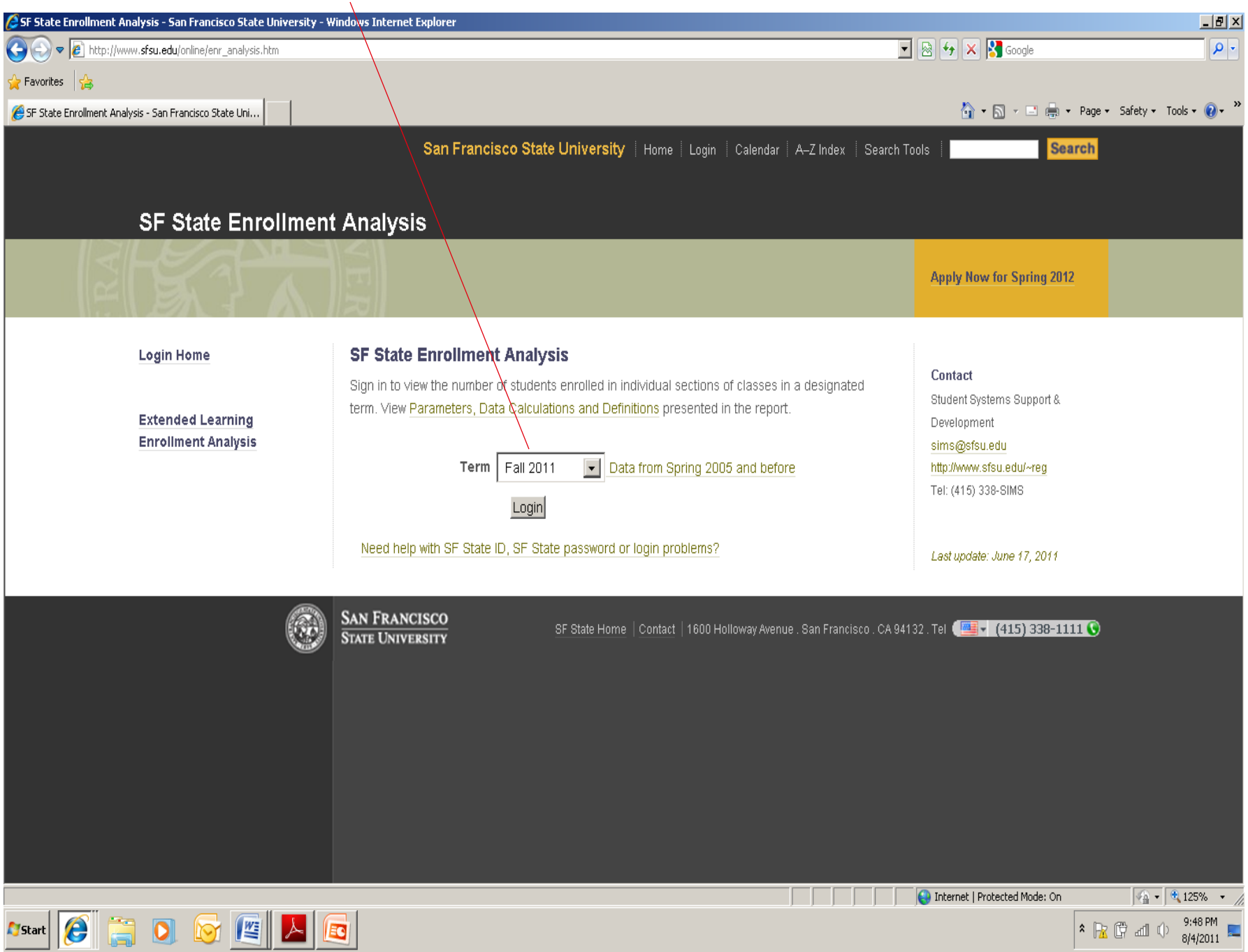
Bad

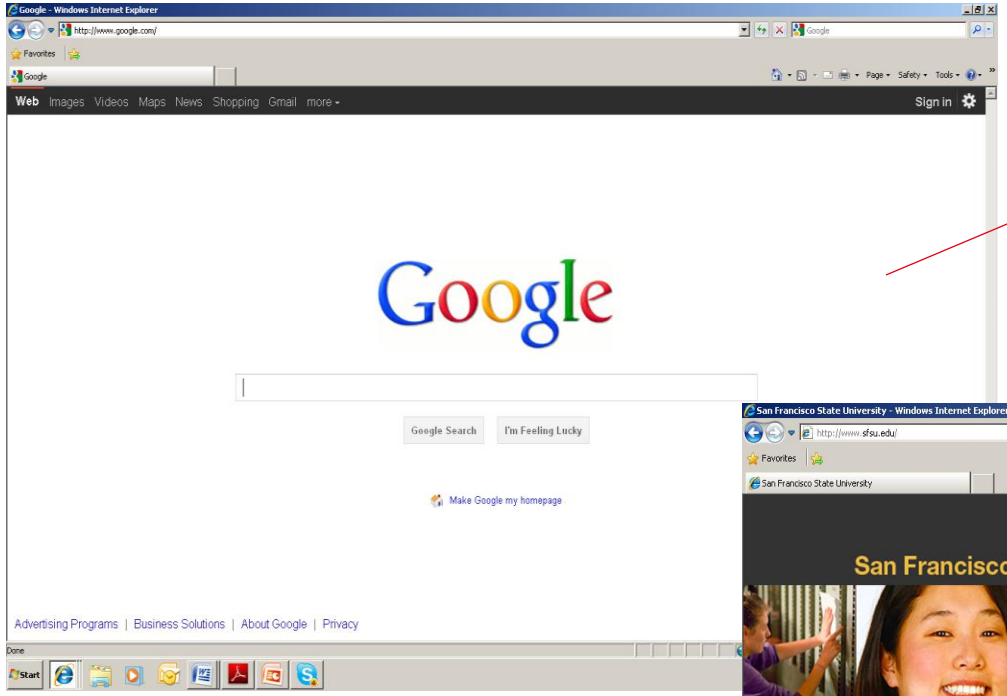
Signing for credit card



What is wrong here?

Asks semester entry before login, login required for each new semester entry

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Sign in
Google
Google Search
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Simplicity

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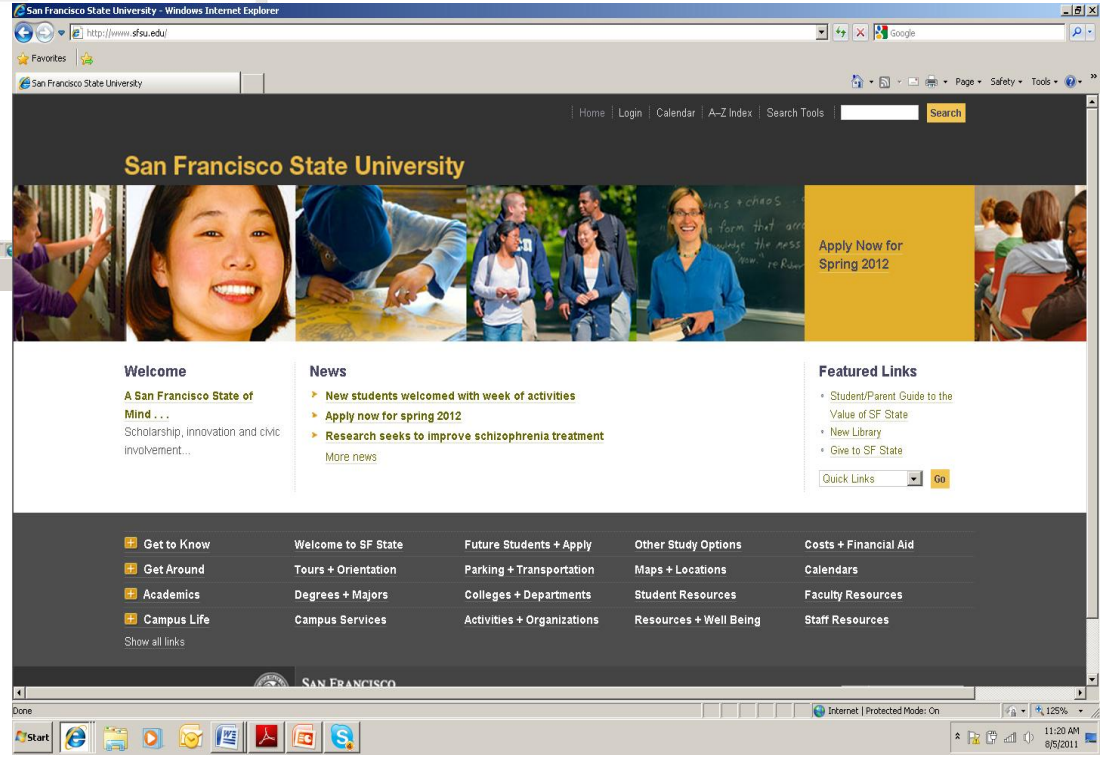
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My profile

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Conflicts of interest	76 last updated May 17, 2016 08:43:33 America/New_York

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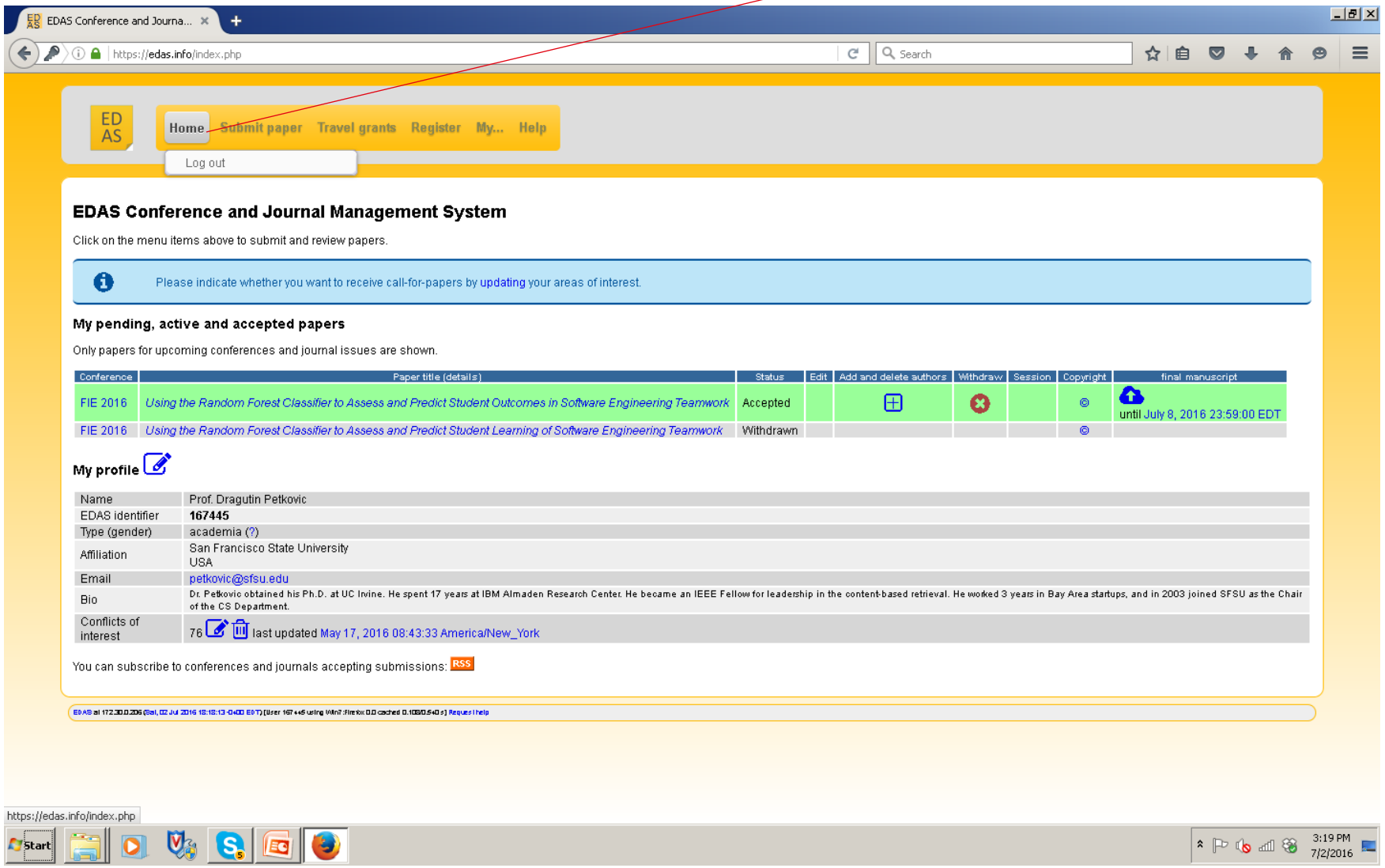
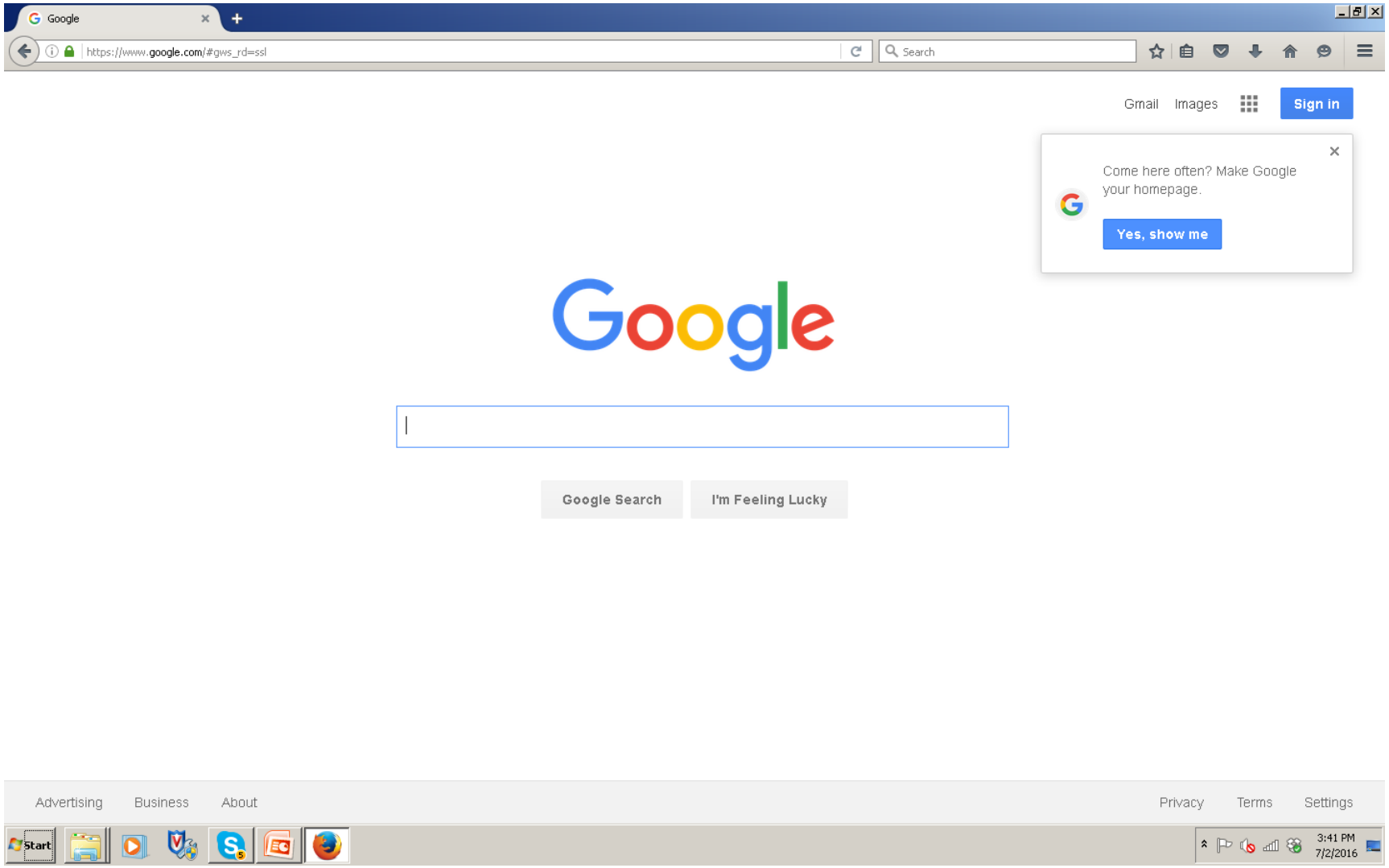


Some good designs - think WHY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Alignment, layout, consistent use of font, help/explanations obvious

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Patent type/status	Return patents with type/status		Any type/status
Date	Return patents anytime Return patents between and e.g. 1999 and 2000, or Jan 1999 and Dec 2000		
Restrict date by	Restrict by filing date Restrict by issue date		

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Design pattern for showing context in browsers –
(**where am I, what I did, what I need to do next**) – use STEP1, 2…
useful when user has to do some steps in predefined order. (From Yahoo)

Also nice grouping into logical units

Clearly says WHAT the apps do, let you TRY it, then says how to START

Google Docs - Online documents, spreadsheets, presentations, surveys, file storage and more - Windows Internet Explorer

https://www.google.com/accounts/ServiceLogin?service=writely&passive=1209600&continue=http://docs.google.com/?hl%3Den%26tab%3Dio&followup=http://docs.google.com/?hl%3D

Google docs

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Excellent search – US News and World Report car guide

Used Nissan Murano for Sale in

https://cars.usnews.com/cars-trucks/used-cars/for-sale/nissan_murano_saratoga_ca?range=75&zip=95070&new_used_type=used&cpo=on&make=Nissan&

Used Nissan Murano near Saratoga, CA

Change Location

Saved Cars 0

Show Results For:

LOCATION: 75 mi of 95070

LISTING TYPE: New; Used; Certified Pre-Owned

VEHICLE HISTORY: CARFAX 1-Owner

MAKE: Nissan

MODEL: Murano

YEAR: 2013 to 2016

Search Inventory

Refine Your Search

PRICE: $0 – $100,000+

MILEAGE: 0 miles – 150,000 miles

Showing 1-20 of 42

Sort by: Relevance

2014 Nissan Murano — $17,000 REDUCED PRICE
in Sunnyvale, CA (6.7 miles away)
Mileage: 38,233
Color: White
Engine: 6 Cyl
Drive: FWD
Transmission: Automatic
See all photos | See More Info | Save
TrueCar Certified Dealer | View the Free CARFAX Report: 1-Owner

2014 Nissan Murano — $17,000
in Modesto, CA (64.4 miles away)
Mileage: 35,376
Color: Silver
Engine: 6 Cyl
Drive: FWD
Transmission: Automatic
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2014 Nissan Murano — $17,548 REDUCED PRICE
in Santa Clara, CA (7.4 miles away)
Mileage: 56,306
Color: Other
Engine: 6 Cyl
Drive: FWD
Transmission: Automatic
See all photos | See More Info | Save
TrueCar Certified Dealer | View the Free CARFAX Report

2014 Nissan Murano — $18,994
in Daly City, CA (37.3 miles away)

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Let us try some search from different providers

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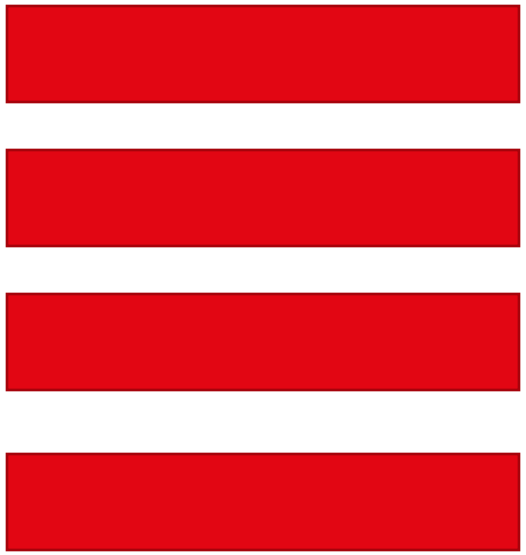
- Which one is more usable?
- Which company is having larger market share?

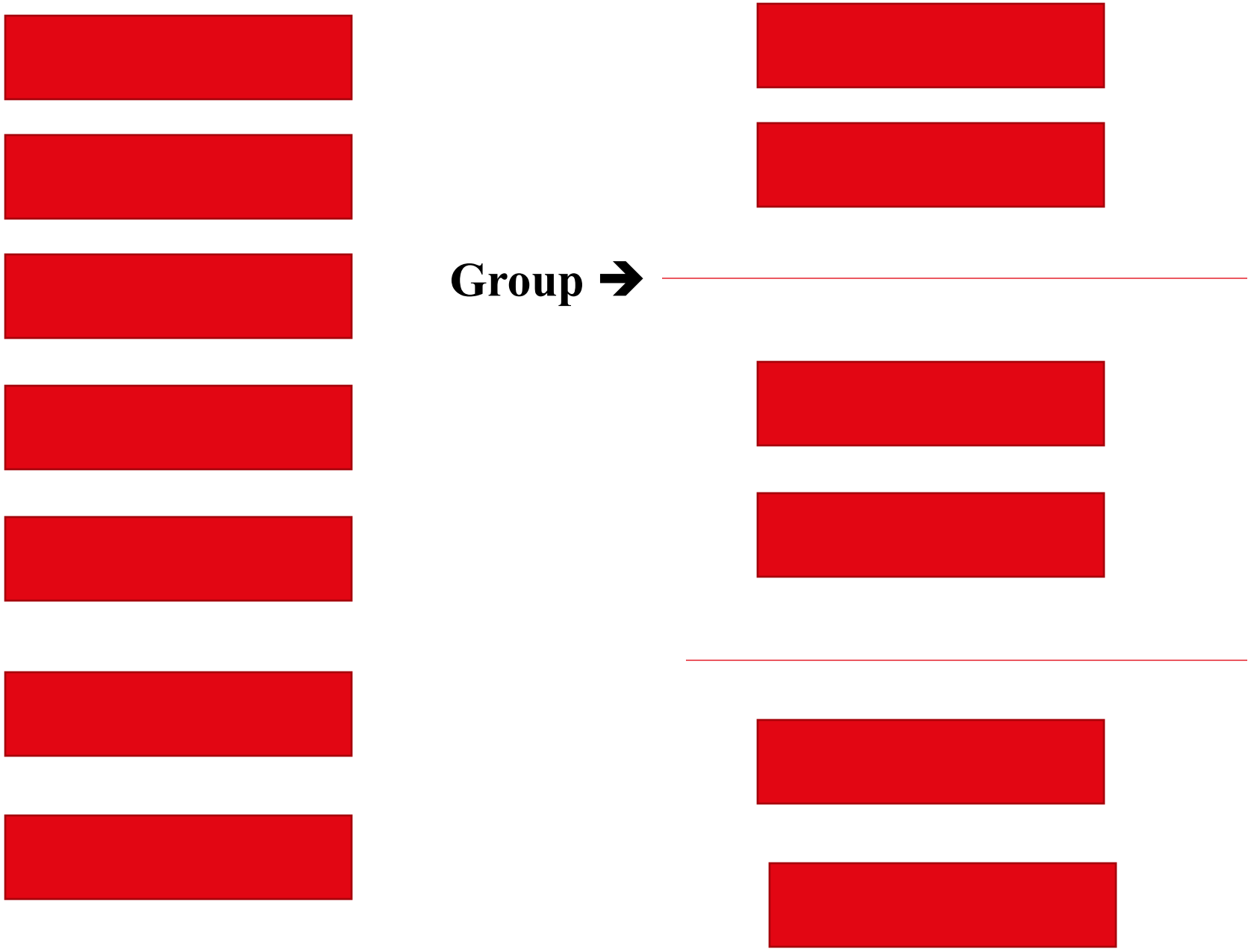
Important

Wrong



Allign ➔ Good

Important



UI Design patterns for specific stuff - FORMS

- Good resources and tools
  - http://mono.company/journal/design-practice/the-10-commandments-of-good-form-design-on-the-web/
  - https://www.nngroup.com/articles/web-form-design
- And of course in bootstrap
  - https://getbootstrap.com/docs/4.3/components/forms/

Use of colors is important

- Colors are part of user experience. They convey:
  - Mood, style
  - Branding (“Big Blue”, Ferrari red, various logos)
  - Aesthetics
  - Consistency (key buttons to have consistent color – Amazon “checkout”)
  - Attention (red vs. green)
  - Ease of reading (good contrast…)

Which colors to chose – tools and guidance available

- http://colormind.io/bootstrap/
- https://material.io/design/color/the-color-system.html#

Google material design

- https://techcrunch.com/2017/04/06/new-google-tool-helps-developers-pick-the-right-colors-for-their-apps/

Apple

- https://developer.apple.com/design/human-interface-guidelines/ios/visual-design/color/

Web safe colors (general)

- https://www.htmlgoodies.com/tutorials/web_graphics/consistent-colors-for-your-site-all-you-need-to-know-about-web-safe-colors.html

Sound as a component of UI and UX

- Sound effects very important in movies, games and UX ➔ integral part of UX
- Some good reads
  - https://uxplanet.org/the-role-of-sounds-in-ux-47adb8f82b38
- Example libraries of sounds – not all free
  - http://raisedbeaches.com/octave/
  - https://www.soundsnap.com/tags/ui

Thank You